Teacher Wu Kuan | See It at a Glance #19 — Two Staircases Make a Rectangle

English lecture notes · YouTube Short fI6JSdNi8Ko

1. The trap question

1 + 2 + … + 100 = ?

2. The method

• Copy the staircase, flip it, join: an n × (n + 1) rectangle.
• Sum = n(n + 1) ÷ 2. For 100: 5050.
• Any arithmetic series: (first + last) × number of terms ÷ 2.

3. Parent check method

Ask your child to draw the 1-to-5 staircase on grid paper and find 15 by halving the rectangle.

Practice

1. 1 + 2 + 3 + … + 20 = ?
2. 1 + 2 + … + 50 = ?
3. 2 + 4 + 6 + … + 100 = ?
4. 1 + 3 + 5 + … + 19 = ?
5. 3 + 6 + 9 + … + 39 = ?

Answers

1. 210
2. 1275
3. 2550 (50 terms)
4. 100 (10 terms)
5. 273 (13 terms)

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